

YOUNGWON KIM

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EDUCATION

- Jun 2023* **Ph.D., Measurement & Statistics**
College of Education, University of Washington, Seattle, WA
Dissertation: *Modern prediction algorithms for factor analyses: Comparing accuracy-efficiency trade-offs* (Advisor: Dr. Elizabeth Sanders)
- May 2017* **M.A., Educational Psychology**
University of Texas at Austin, Austin, TX
Thesis: *A clash of constructs? Re-examining grit in light of academic buoyancy and future time perspective* (Advisor: Dr. Diane Schallert)
- Feb 2011* **B.A., International & Area Studies, Counseling Psychology, & Social Welfare**
Handong Global University, Pohang, Gyeongbuk, South Korea

RESEARCH INTERESTS

- Clinical AI: Foundation models, large language models (LLMs), and machine learning for predictive modeling with electronic health records (EHRs)
- Evaluation and Measurement: Validity, reliability, and benchmark design for AI
- Causal Inference: Randomized controlled trials and observational studies
- Human Behavior & Society: Psychology, mental/physical health, social determinants

RESEARCH EXPERIENCE

- Oct 2024 – Present* **Intelligence ML, Truveta, Bellevue, WA**
Machine Learning Postdoctoral Researcher
- Trained oncology-specific and general-purpose clinical foundation models on longitudinal EHR data to predict patient health trajectories
 - Post-trained generative models with reinforcement learning to align simulated trajectories with real-world patient data
 - Constructed benchmark datasets and a multi-level evaluation framework for clinical foundation models and LLM agents, ranging from low- and mid-level prediction tasks to simulated real-world clinical cases, with attention to calibration, interpretability, fairness, and clinical validity
 - Developed longitudinal ML and LLM risk prediction models for oncology and heart failure, including EHR-based estimation of patient-reported outcomes such as Kansas City Cardiomyopathy Questionnaire scores
 - Designed Model Context Protocol (MCP) tools and deep-agent systems integrating predictive models with interactive dashboards, automated prompt optimization, and agentic clinical workflows
- Aug 2023 – Sep 2024* **Causality, Applications, and Research in Education and Statistics, Harvard Graduate School of Education, Harvard University, Cambridge, MA**
Postdoctoral Fellow
- Project:** *Practical tools for large-scale evaluation of text data in randomized trials in education* (Drs. Luke Miratrix, Reagan Mozer, and Shireen Al-Adeimi)
- Conducted research on the performance and accuracy of large language models versus machine learning algorithms for automated essay scoring
 - Developed [rectext](#), an R package for analyzing text outcomes in randomized controlled trials (RCTs)
 - Created academic papers, tutorial materials, and blog posts to assist applied researchers in utilizing text in RCT contexts
- Jun 2019 – Jun 2023* **Center for Social Science Computation and Research, University of Washington, Seattle, WA (Dr. Jerald Herting)**
Statistical Consultant
- Provided statistical and programming consulting services to students and faculty from across UW Seattle campus
 - Offered guidance on research designs, data cleaning, data analysis, and use of data analysis and visualization software (*Mplus, Python, R, SPSS, SQL, and Tableau*)
 - Conducted remote synchronous and asynchronous workshops on *R* and *SPSS*

RESEARCH EXPERIENCE (Cont'd)

- Jun 2022 – Aug 2022* **Human Services Division - Education Systems Program Area, American Institutes for Research (AIR), Arlington, VA (Dr. Cong Ye)**
- Developed a school-level social vulnerability index for the CEE (COVID-19 and Equity in Education) projects
 - Participated in AIR Hackathon using the CEE longitudinal database and 2022 Summer Intern Expo
- May 2018 – Jun 2022* **School of Education & Human Development, Texas A&M University, College Station, TX**
- Project:** *Asian international graduate students' professional development and psychological well-being* (Dr. Katie Koo)
- Interviewed international graduate students and transcribed the interviews
 - Coded and analyzed transcripts on the usage of social media, language proficiency, academic achievement, career development, social support, mental and physical health issues of international students
 - Contributed to writing proposals for conference presentations and manuscripts for publication
- Mar 2020 – Dec 2020* **Center for the Study of Teaching and Policy, University of Washington, Seattle, WA**
- Project:** *Teacher workforce* (Drs. Ana Elfers and Marge Plecki)
- Co-designed and implemented statistical models to examine the teacher workforce in Washington State with respect to teacher retention and mobility
 - Co-authored results for presentation, white paper, and publication
- Jul 2018 – Sep 2018* **Health Sciences Library, University of Washington, Seattle, WA (Dr. Lorrie Dong)**
- Performed data cleaning and created dashboards for National Network of Libraries of Medicine (NNLM)
 - Created R functions to automate data-cleaning processes
- Feb 2015 – Aug 2017* **Higher Education Learning and Persistence Lab, College of Education, University of Texas at Austin, Austin, TX (Dr. Carlton Fong)**
- Collected documents and coded data for meta-analyses on psychosocial factors and critical thinking of community college students
 - Conducted meta-analyses to examine growth- and fixed-mindset profiles and associations with stress, need satisfaction, coping strategies, and fear of failure

TEACHING EXPERIENCE

University of Washington

- Sep 2021 – Dec 2021* **EDPSY 576: Multilevel Modeling**, College of Education
- Apr 2019 – Jun 2019* **PSYCH 315: Understanding Statistics in Psychology**, Department of Psychology
- Sep 2018 – Mar 2019* **PSYCH 209: Fundamentals of Psychological Research**, Department of Psychology
- Sep 2017 – Dec 2017* **EDPSY 302: Child Development and Learning**, College of Education

Handong Global University

- Jul 2020 & Jul 2021* **Online Summer Workshop on Applied Statistics**, School of Counseling Psychology and Social Welfare

University of Texas at Austin

- Jan 2017 – May 2017* **SDS 332: Statistical Models for the Health and Behavioral Sciences**, Department of Statistics and Data Sciences
- Sep 2015 – Dec 2016* **MKT 282: Marketing Analytics & Applications**, McCombs School of Business

PUBLICATIONS

Peer-reviewed articles

- Zeng, S., **Kim, Y. W.**, Lau, W., Alipour, E., Etzioni, R., Yetisgen, M., & Oka, A. (in press). TrajOnco: A multi-agent framework for temporal reasoning over longitudinal EHR for multi-cancer early detection. *npj Digital Medicine*. <https://doi.org/10.48550/arXiv.2604.10386>
- Fong, C. J., Schallert, D. L., Williamson, Z. H., Lin, S., Williams, K. M., & **Kim, Y. W.** (2023). Are self-compassionate writers more feedback literate? Exploring undergraduates' perceptions of feedback constructiveness. *Assessing Writing*, 57, Article 100761. <https://doi.org/10.1016/j.asw.2023.100761>
- Elfers, A., Plecki, M., **Kim, Y. W.**, & Bei, N. (2022). How retention and mobility rates differ for teachers of colour. *Teachers and Teaching*, 28(6), 724–741. <https://doi.org/10.1080/13540602.2022.2098271>
- Fong, C. J., Schallert, D. L., Williams, K. M., Williamson, Z. H., Lin, S., **Kim, Y. W.**, & Chen, L. (2021). Making feedback constructive: The interplay of undergraduates' motivation with perceptions of feedback specificity and friendliness. *Educational Psychology*, 41(10), 1241–1259. <https://doi.org/10.1080/01443410.2021.1951671>
- Fong, C. J., & **Kim, Y. W.** (2021). A clash of constructs? Re-examining grit in light of academic buoyancy and future time perspective. *Current Psychology*, 40(4), 1824–1837. <https://doi.org/10.1007/s12144-018-0120-4>
- Koo, K., **Kim, Y. W.**, Lee, J., & Nyunt, G. (2021). “It’s my fault”: Exploring experiences and mental wellness among Korean international graduate students. *Journal of International Students*, 11(4), 790–811. <https://doi.org/10.32674/jis.v11i4.2801>
- Fong, C. J., Schallert, D. L., Williams, K. M., Williamson, Z. H., Warner, J. R., Lin, S., & **Kim, Y. W.** (2018). When feedback signals failure but offers hope for improvement: A process model of constructive criticism. *Thinking Skills and Creativity*, 30, 42–53. <https://doi.org/10.1016/j.tsc.2018.02.014>
- Fong, C. J., Williams, K. M., Williamson, Z. H., Lin, S., **Kim, Y. W.**, & Schallert, D. L. (2018). “Inside out”: Appraisals for achievement emotions from constructive, positive, and negative feedback on writing. *Motivation and Emotion*, 42, 236–257. <https://doi.org/10.1007/s11031-017-9658-y>
- Fong, C. J., Davis, C. W., Kim, Y., **Kim, Y. W.**, Marriott, L. A., & Kim, S. (2017). Psychosocial factors and community college student success: A meta-analytic investigation. *Review of Educational Research*, 87(2), 388–424. <https://doi.org/10.3102/0034654316653479>
- Fong, C. J., Kim, Y., Davis, C. W., Hoang, T., & **Kim, Y. W.** (2017). A meta-analysis on critical thinking and community college achievement. *Thinking Skills and Creativity*, 26, 71–83. <https://doi.org/10.1016/j.tsc.2017.06.002>

Working papers

- Kim, Y. W.**, Mozer, R., Al-Adeimi, S., & Miratrix, L. (2025). *ChatGPT vs. machine learning: Assessing the efficacy and accuracy of large language models for automated essay scoring* (EdWorkingPaper No. 25-1335). Annenberg Institute at Brown University. <https://doi.org/10.26300/7vj9-5y53>

Peer-reviewed conference proceedings

- Lau, W., **Kim, Y.**, Parasa, S., Haque, M. E., Oka, A., & Nanduri, J. (2026). Predicting early-onset colorectal cancer with large language models. *AMIA Annual Symposium Proceedings*, 2025, 653–662. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12919599/>

PUBLICATIONS (Cont'd)

Published conference abstracts

- Kim, Y.**, Lau, W., Alipour, E., Zeng, S., Oka, A., & Nanduri, J. (2026). Characterizing oncology patient journeys and health state transitions using a data-driven Markov transition matrix in large-scale electronic health records. *Value in Health*, 29(6, Suppl. 1), S309–S310. <https://doi.org/10.1016/j.jval.2026.03.1567>
- Zeng, S., **Kim, Y.**, Lau, W., Alipour, E., Etzioni, R., Yetisgen, M., Oka, A., & Nanduri, J. (2026). Zero-shot lung cancer risk prediction from longitudinal electronic health records with chain-of-agents framework. *Value in Health*, 29(6, Suppl. 1), S309. <https://doi.org/10.1016/j.jval.2026.03.1566>
- Lau, W., Alipour, E., **Kim, Y.**, Zeng, S., Oka, A., & Nanduri, J. (2026). Can a generative patient journey foundation model alleviate the burden of cancer screening? *Value in Health*, 29(6, Suppl. 1), S333. <https://doi.org/10.1016/j.jval.2026.03.1671>
- Alipour, E., Lau, W., **Kim, Y.**, Zeng, S., Oka, A., & Nanduri, J. (2026). Patient journey foundational model for scalable imputation of missing units of measurement (UOM) in electronic health records data. *Value in Health*, 29(6, Suppl. 1), S430. <https://doi.org/10.1016/j.jval.2026.03.2128>
- Zeng, S., **Kim, Y.**, Lau, W., Alipour, E., Etzioni, R. D., Yetisgen, M., Oka, A., & Nanduri, J. (2026). Predicting multi-cancer risk from EHR data using multi-agent LLMs. *Journal of Clinical Oncology*, 44(16_suppl), 10501. https://doi.org/10.1200/JCO.2026.44.16_suppl.10501
- Lau, W., **Kim, Y.**, Parasa, S., Haque, M. E., Daraei, S., Rao, R., Pillai, J., & Oka, A. (2025). Development and validation of machine learning risk prediction models for detection of early-onset colorectal cancer: Data from 30 health systems in the United States. *Journal of Clinical Oncology*, 43(16_suppl), 3613. https://doi.org/10.1200/JCO.2025.43.16_suppl.3613

Preprints

- Kim, Y. W.**, Lau, W., Patel, N., Kendrick, K., Wu, A., Feldman, T., Ahern, R., & Oka, A. (2026). *Electronic health record-based estimation of Kansas City Cardiomyopathy Questionnaire scores in heart failure*. medRxiv. <https://doi.org/10.64898/2026.04.03.26350138>

CONFERENCE PRESENTATIONS

International Society for Pharmacoeconomics and Outcomes Research (ISPOR)

- Kim, Y.**, Lau, W., Alipour, E., Zeng, S., Oka, A., & Nanduri, J. (2026, May). *Characterizing oncology patient journeys and health state transitions using a data-driven Markov transition matrix in large-scale electronic health records*. Poster presented at the ISPOR Annual Meeting, Philadelphia, PA.
- Zeng, S., **Kim, Y.**, Lau, W., Alipour, E., Etzioni, R., Yetisgen, M., Oka, A., & Nanduri, J. (2026, May). *Zero-shot lung cancer risk prediction from longitudinal electronic health records with chain-of-agents framework*. Poster presented at the ISPOR Annual Meeting, Philadelphia, PA.
- Lau, W., Alipour, E., **Kim, Y.**, Zeng, S., Oka, A., & Nanduri, J. (2026, May). *Can a generative patient journey foundation model alleviate the burden of cancer screening?* Poster presented at the ISPOR Annual Meeting, Philadelphia, PA.
- Alipour, E., Lau, W., **Kim, Y.**, Zeng, S., Oka, A., & Nanduri, J. (2026, May). *Patient journey foundation model for scalable imputation of missing units of measurement in electronic health records data*. Poster presented at the ISPOR Annual Meeting, Philadelphia, PA.

American Medical Informatics Association (AMIA)

- Lau, W., **Kim, Y.**, Parasa, S., Haque, M. E., Oka, A., & Nanduri, J. (2025, November). *Predicting early-onset colorectal cancer with large language models*. Paper presented at the AMIA Annual Symposium, Atlanta, GA.

CONFERENCE PRESENTATIONS (Cont'd)

American Society of Clinical Oncology (ASCO)

Zeng, S., **Kim, Y.**, Lau, W., Alipour, E., Etzioni, R. D., Yetisgen, M., Oka, A., & Nanduri, J. (2026, May). Predicting multi-cancer risk from EHR data using multi-agent LLMs. Oral presentation at the ASCO Annual Meeting, Chicago, IL.

Lau, W., **Kim, Y.**, Parasa, S., Haque, M. E., Daraei, S., Rao, R., Pillai, J., & Oka, A. (2025, May). *Development and validation of machine learning risk prediction models for detection of early-onset colorectal cancer: Data from 30 health systems in the United States*. Poster presented at the ASCO Annual Meeting, Chicago, IL.

Joint Statistical Meetings (JSM)

Kim, Y., Mozer, R., Miratrix, L., & Al-Adeimi, S. (2024, August). *ChatGPT vs. machine learning: Assessing the efficacy and accuracy of large language models for automated essay scoring*. Paper presented at the JSM Annual Meeting, Portland, OR.

Association for Education Finance & Policy (AEFP)

Kim, Y., Mozer, R., Miratrix, L., & Al-Adeimi, S. (2024, March). *Assessing essay grading with machine learning and large language models: A comparative study on efficacy and accuracy*. Paper presented at the AEFP Annual Meeting, Baltimore, MD.

National Council on Measurement in Education (NCME)

Kim, Y., & Sanders, E. A. (2023, April). *Exploratory measurement modeling with Lasso*. Paper presented at the NCME Annual Meeting, Chicago, IL.

Bei, N., Sanders, E. A., Abe, N., & **Kim, Y.** (2021, June). *Measuring item process data with network analysis methods*. Paper presented at the NCME Annual Meeting (Virtual).

International Meeting of the Psychometric Society (IMPS)

Kim, Y., & Sanders, E. A. (2023, July). *Exploratory measurement modeling with Lasso*. Paper presented at the IMPS Annual Meeting, College Park, MD.

American Educational Research Association (AERA)

Kim, Y., & Sanders, E. A. (2024, April). *Enhancing factor analysis: Evaluating factor retention accuracy with measurement error*. Paper presented at the AERA Annual Meeting, Philadelphia, PA.

Koo, K., Kwon, J., & **Kim, Y.** (2022, April). *Strive to survive in a pandemic in a foreign country: Korean international students' well-being during COVID-19*. Paper presented at the AERA Annual Meeting, San Diego, CA.

Kim, Y. (2021, April). *Understanding the role of academic achievement and psychosocial factors on adolescent depression: A latent growth model*. Paper presented at the AERA Annual Meeting (Virtual).

Koo, K., **Kim, Y.**, Lee, J., & Rodriguez, S. (2021, April). *What does it mean to be an Asian in a pandemic? Exploring experiences of racism and discrimination among Asian international students in the United States during COVID-19*. Paper presented at the AERA Annual Meeting (Virtual).

Elfers, A., Plecki, M., **Kim, Y.**, & Bei, N. (2021, April). *Examining differences in the retention and mobility of teachers of color by racial/ethnic group*. Paper presented at the AERA Annual Meeting (Virtual).

Kim, Y. (2020, April). *Executive function difficulty, self-esteem, happiness, gender, and parent's income as the predictors of digital addiction*. Paper accepted for presentation at the AERA Annual Meeting, San Francisco, CA. (Conference canceled)

CONFERENCE PRESENTATIONS (Cont'd)

American Educational Research Association (AERA, Cont'd)

- Koo, K., **Kim, Y.**, Lee, J., & Nyunt, G. (2020, April). *What makes them happy? A qualitative study on Korean international graduate students' experiences and psychological well-being*. Paper accepted for presentation at the AERA Annual Meeting, San Francisco, CA. (Conference canceled)
- Kim, Y.** (2019, April). *Trajectories of parents' socioeconomic status, child-parent interaction, children's self-control, school readiness, and academic performance*. Paper presented at the AERA Annual Meeting, Toronto, ON.
- Fong, C. J., Davis, C., Li, D., & **Kim, Y.** (2017, April). *A meta-analysis of noncognitive factors and community college student success at 4-year institutions*. Paper presented at the AERA Annual Meeting, San Antonio, TX.
- Fong, C. J., **Kim, Y.**, Lin, S., & Hoang, T. (2017, April). *Do noncognitive interventions enhance community college student success? A systematic review and meta-analysis*. Paper presented at the AERA Annual Meeting, San Antonio, TX.
- Williamson, Z. H., Fong, C. J., Lin, S., Williams, K. M., **Kim, Y.**, & Schallert, D. L. (2017, April). *Self-kindness in the face of criticism: Self-compassion influences undergraduates' perceptions of constructive feedback*. Paper presented at the AERA Annual Meeting, San Antonio, TX.
- Fong, C. J., Kim, Y., Davis, C. W., **Kim, Y.**, Marriott, L. A., & Kim, S. (2016, April). *Psychosocial and self-regulatory correlates of community college student success: A systematic review and meta-analysis*. Paper presented at the AERA Annual Meeting, Washington, D.C.
- Fong, C. J., Schallert, D. L., Williams, K. M., Williamson, Z. H., Lin, S., Chen, L., & **Kim, Y.** (2016, April). *Making feedback constructive: The interplay of undergraduates' mastery goal orientation with feedback specificity and friendliness*. Paper presented at the AERA Annual Meeting, Washington, D.C.

Korean-American Educational Researchers Association (KAERA)

- Kim, Y.** (2020, April). *Understanding the role of academic achievement and psychosocial factors on adolescent depression: A latent growth model*. Paper accepted for presentation at the KAERA Annual Meeting, San Francisco, CA. (Conference canceled)
- Koo, K., & **Kim, Y.** (2020, April). *The effectiveness of focusing art therapy on PTSD and cultural adjustment among North Korean refugee girls*. Paper accepted for presentation at the KAERA Annual Meeting, San Francisco, CA. (Conference canceled)
- Koo, K., **Kim, Y.**, & Lee, J. (2019, April). *What makes them happy? Korean international graduate students' personal and professional experiences and emotional health*. Paper presented at the KAERA Annual Meeting, Toronto, ON.

Other Refereed Conferences

- Koo, K., **Kim, Y.**, & Lee, J. (2019, March). *What makes them happy? Asian international graduate students' personal and professional experiences and emotional health*. Paper presented at the ACPA Annual Meeting, Boston, MA.
- Fong, C. J., Davis, C., Li, D., **Kim, Y.**, & Hartman, C. E. (2016, August). *A meta-analysis of noncognitive factors and community college transfer student success at four-year institutions*. Paper presented at the Texas Higher Education Symposium, Austin, TX.
- Lin, S., Fong, C. J., & **Kim, Y.** (2016, April). *Make up your mind: A person-centered approach to growth and fixed mindsets among Chinese undergraduates*. Paper presented at the Chinese American Educational Research & Development Association, Washington, D.C.
- Kim, Y., Fong, C. J., Davis, C. W., Hoang, T., & **Kim, Y.** (2016, March). *A meta-analysis on critical thinking and community college student achievement*. Paper presented at the Council for the Study of Community Colleges Annual Meeting, Plano, TX.

INVITED TALKS AND PRESENTATIONS

- Kim, Y.** (2026, May). *Research with healthcare databases* [Invited talk]. AI in Therapeutic Science, Gachon University, South Korea (Virtual).
- Kim, Y.** (2026, January). *Clinical AI research with large-scale healthcare data* [Invited talk]. Healthcare Simulation and Optimization Lab, Incheon National University, South Korea (Virtual).
- Kim, Y.** (2024, November). *Large language model research in education* [Invited talk]. VIP Data-Driven Education program, Georgia Institute of Technology, Atlanta, GA (Virtual).
- Kim, Y.** (2023, October). *Modern prediction algorithms for factor analyses: Comparing accuracy-efficiency trade-offs* [Invited talk]. Mathematical Sciences Research Seminar, Bentley University, Waltham, MA ([Virtual](#)).
- Kim, Y.** (2023, October). *Enhancing factor analysis: Evaluating factor retention accuracy with measurement error* [Invited talk]. CARES Lab (Causality, Applications, and Research in Education and Statistics), Harvard Graduate School of Education / Harvard Department of Statistics, Harvard University, Cambridge, MA.
- Kim, Y.** (2021, April). *Racism and the racialized experience among Asian international students in the U.S.* [Invited panel]. ACPA 2021 International Colloquium Support for international students during and after COVID-19: Best practices from a global perspective ([Virtual](#)).
- Kim, Y.** (2020, July). *Racism and racialized identities: The experience among Asian international students in the United States* [Invited panel]. ACPA Commission for Global Dimensions of Student Development (Virtual).

SERVICE

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| <i>Jan 2023 – Present</i>
PR Team Member | <ul style="list-style-type: none"> Changbal, Korean-American Tech Community, Seattle, WA |
| <i>Oct 2020 – Present</i>
Ad-hoc Reviewer | <ul style="list-style-type: none"> Journal of International Students, Current Psychology, AMIA Annual Symposium |
| <i>Oct 2022</i>
Program Planning Committee Member | <ul style="list-style-type: none"> Section 2 (Statistical theory and quantitative methodologies), Division D program planning committee meeting, AERA, Washington, D.C. |
| <i>Aug 2021 – Aug 2022</i>
Student Support Chair | <ul style="list-style-type: none"> Associated Students of the College of Education (ASCE), University of Washington, Seattle |
| <i>Aug 2021 – Aug 2022</i>
GSC Liaison | <ul style="list-style-type: none"> AERA Division D Graduate Student Committee (GSC) Liaison, American Educational Research Association, Washington, D.C. |
| <i>Apr 2021 – Jun 2021</i>
Mentor | <ul style="list-style-type: none"> Next Generation Mentorship Program (NGMP), Office of Student Diversity and Inclusion, College of Education, University of Washington, Seattle |
| <i>Jul 2019 – Jan 2020</i>
Student Representative | <ul style="list-style-type: none"> New Hire Search Committee, Measurement & Statistics, College of Education, University of Washington, Seattle |

AWARDS/HONORS

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| 2019 | Young Researcher Award, Korean-American Educational Researchers Association (KAERA) |
| 2015 – 2021 | OTEFE Foundation Scholarship, OTEFE Foundation, Inc. |